

gas2 Module — Thermodynamic Potentials and Phase Equilibrium

Theoretical Foundation Document

VSEPR-SIM Development

Version 1.0.0 - 4.X Beta — April 2026

Contents

1 Scope and Purpose

This document establishes the theoretical foundation for the thermodynamic potential infrastructure in the gas2 module. The goal is to compute the four fundamental potentials

$$A(T, V, N), \quad G(T, P, N), \quad \mu_i(T, P, \mathbf{x}), \quad S(T, V, N)$$

from a partition-function basis, then derive phase boundaries from rigorous equilibrium conditions rather than empirical curve fits.

1.1 Design Principles

1. **SI internally, Hartree at render time.** All thermodynamic quantities are stored in SI (J, Pa, m³, K). Hartree conversion is applied at the display layer only:

$$E_{\text{Eh}} = \frac{E_{\text{J}}}{E_{\text{h}}}, \quad E_{\text{h}} = 4.3597447222071 \times 10^{-18} \text{ J}$$

2. **Anti-black-box.** Every intermediate—partition function, fugacity coefficient, chemical potential, activity coefficient—is stored in the result struct and available for inspection.
3. **Deterministic.** Given $(T, P, N, \text{species})$, outputs are reproducible with no hidden stochastic state.

2 Partition Function and Free Energy

2.1 Classical Partition Function

For an N -particle system with Hamiltonian $H(\mathbf{q}, \mathbf{p})$, the classical canonical partition function is

$$Z(T, V, N) = \frac{1}{N! h^{3N}} \int e^{-\beta H(\mathbf{q}, \mathbf{p})} d\mathbf{q} d\mathbf{p}, \quad \beta = \frac{1}{k_B T}. \quad (1)$$

2.2 Ideal Gas Partition Function

For a single-component ideal gas of N molecules with molar mass M , the translational partition function factorises:

$$Z_{\text{trans}} = \frac{V^N}{N! \Lambda^{3N}}, \quad \Lambda = \sqrt{\frac{h^2}{2\pi m k_B T}} \quad (2)$$

where Λ is the thermal de Broglie wavelength and $m = M/N_A$.

Internal degrees of freedom contribute multiplicatively:

$$Z = Z_{\text{trans}} \cdot Z_{\text{rot}}^N \cdot Z_{\text{vib}}^N \cdot Z_{\text{elec}}^N. \quad (3)$$

For a rigid rotor:

$$Z_{\text{rot}} = \begin{cases} 1, & \text{monoatomic} \\ \frac{T}{\sigma_{\text{rot}} \Theta_{\text{rot}}}, & \text{linear, } \Theta_{\text{rot}} = \frac{h^2}{2Ik_B} \\ \frac{\sqrt{\pi}}{\sigma_{\text{rot}}} \left(\frac{T}{\Theta_A}\right)^{1/2} \left(\frac{T}{\Theta_B}\right)^{1/2} \left(\frac{T}{\Theta_C}\right)^{1/2}, & \text{nonlinear} \end{cases} \quad (4)$$

For a harmonic oscillator mode k :

$$Z_{\text{vib},k} = \frac{e^{-\Theta_{v,k}/(2T)}}{1 - e^{-\Theta_{v,k}/T}}, \quad \Theta_{v,k} = \frac{h\nu_k}{k_B}. \quad (5)$$

3 Thermodynamic Potentials

3.1 Helmholtz Free Energy $A(T, V, N)$

The Helmholtz free energy follows directly from the partition function:

$$A(T, V, N) = -k_B T \ln Z(T, V, N). \quad (6)$$

For the ideal gas:

$$A^{\text{ig}}(T, V, N) = -Nk_B T \left[\ln \left(\frac{V}{N\Lambda^3} \right) + 1 \right] - Nk_B T \ln Z_{\text{int}}(T). \quad (7)$$

For a real gas described by a cubic equation of state (Van der Waals, RK, PR), the departure function is

$$A^{\text{dep}}(T, V, N) = A(T, V, N) - A^{\text{ig}}(T, V, N) = - \int_{\infty}^V \left(P - \frac{NRT}{V'} \right) dV'. \quad (8)$$

Van der Waals departure:

$$A_{\text{VdW}}^{\text{dep}} = -NRT \ln \left(\frac{V - Nb}{V} \right) - \frac{N^2 a}{V}. \quad (9)$$

Redlich–Kwong departure:

$$A_{\text{RK}}^{\text{dep}} = -NRT \ln \left(\frac{V - Nb}{V} \right) - \frac{Na}{b\sqrt{T}} \ln \left(\frac{V + Nb}{V} \right). \quad (10)$$

3.2 Gibbs Free Energy $G(T, P, N)$

The Gibbs free energy is obtained via Legendre transform:

$$G(T, P, N) = A(T, V, N) + PV. \quad (11)$$

In practice, we solve for $V(T, P)$ from the EOS, then evaluate:

$$G = A^{\text{ig}}(T, V, N) + A^{\text{dep}}(T, V, N) + PV. \quad (12)$$

3.3 Entropy

$$S(T, V, N) = - \left(\frac{\partial A}{\partial T} \right)_{V, N}. \quad (13)$$

For the ideal gas (Sackur–Tetrode):

$$S^{\text{ig}} = Nk_B \left[\frac{5}{2} + \ln \left(\frac{V}{N\Lambda^3} \right) \right] + S_{\text{int}}(T). \quad (14)$$

4 Chemical Potential and Phase Equilibrium

4.1 Chemical Potential $\mu_i(T, P, \mathbf{x})$

For component i in a mixture:

$$\mu_i(T, P, \mathbf{x}) = \mu_i^\circ(T) + RT \ln \left(\frac{f_i}{f_i^\circ} \right), \quad (15)$$

where f_i is the fugacity of component i and f_i° is the reference fugacity (typically 1 bar for gases).

The fugacity coefficient from a cubic EOS:

$$\ln \phi_i = \frac{1}{RT} \int_V^\infty \left[\left(\frac{\partial P}{\partial N_i} \right)_{T, V, N_{j \neq i}} - \frac{RT}{V} \right] dV - \ln Z. \quad (16)$$

4.2 Phase Equilibrium Conditions

At phase equilibrium between phases α and β :

$$T^\alpha = T^\beta, \quad (17)$$

$$P^\alpha = P^\beta, \quad (18)$$

$$\mu_i^\alpha(T, P, \mathbf{x}^\alpha) = \mu_i^\beta(T, P, \mathbf{x}^\beta) \quad \forall i. \quad (19)$$

For a pure substance, this reduces to finding (T, P) where the Gibbs free energy of the two phases is equal:

$$G^\alpha(T, P) = G^\beta(T, P). \quad (20)$$

4.3 Maxwell Construction (Equal-Area Rule)

For a cubic EOS at subcritical temperatures, the pressure–volume isotherm exhibits a van der Waals loop. The saturation pressure P_{sat} is determined by the equal-area condition:

$$\int_{V_{\text{liq}}}^{V_{\text{vap}}} [P(V) - P_{\text{sat}}] dV = 0, \quad (21)$$

which is equivalent to requiring $G^{\text{liq}} = G^{\text{vap}}$.

This is implemented as a root-finding problem: find P_{sat} such that the fugacities of the liquid and vapour roots are equal.

5 Intermolecular Potential Decomposition

The total potential energy of a molecular system is decomposed as:

$$U_{\text{total}} = U_{\text{bond}} + U_{\text{angle}} + U_{\text{torsion}} + U_{\text{vdW}} + U_{\text{Coul}} + U_{\text{pol}} + U_{\text{many-body}}. \quad (22)$$

5.1 Bonded Interactions

Bond stretching (harmonic):

$$U_{\text{bond}} = \sum_{\text{bonds}} \frac{1}{2} k_r (r - r_0)^2. \quad (23)$$

Angle bending (harmonic):

$$U_{\text{angle}} = \sum_{\text{angles}} \frac{1}{2} k_{\theta} (\theta - \theta_0)^2. \quad (24)$$

Torsional rotation:

$$U_{\text{torsion}} = \sum_{\text{dihedrals}} \frac{V_n}{2} [1 + \cos(n\phi - \phi_0)]. \quad (25)$$

5.2 Non-bonded Interactions

Van der Waals (Lennard-Jones 12-6):

$$U_{\text{vdW}} = \sum_{i < j} 4\epsilon_{ij} \left[\left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}} \right)^6 \right]. \quad (26)$$

Coulomb electrostatics:

$$U_{\text{Coul}} = \sum_{i < j} \frac{k_e q_i q_j}{r_{ij}}. \quad (27)$$

Polarization (induced dipole, SCF):

$$U_{\text{pol}} = -\frac{1}{2} \sum_i \boldsymbol{\mu}_i^{\text{ind}} \cdot \mathbf{E}_i^{\text{perm}}, \quad (28)$$

where $\boldsymbol{\mu}_i^{\text{ind}} = \alpha_i \mathbf{E}_i^{\text{total}}$ and the total field is solved self-consistently (see `polarization_scf.cpp`).

Many-body corrections:

$$U_{\text{many-body}} = U_{\text{3-body}} + U_{\text{association}} + \dots \quad (29)$$

These include Axilrod–Teller–Muto triple-dipole terms and hydrogen-bonding association models where applicable.

6 Coarse-Grained Free-Energy Functional

Near a phase transition, the relevant physics is captured by a density order parameter:

$$\phi(\mathbf{r}) = \rho(\mathbf{r}) - \rho_c, \quad (30)$$

where ρ_c is the critical density.

The Landau–Ginzburg free-energy functional:

$$F[\phi] = \int [a \phi^2 + b \phi^4 + \kappa (\nabla \phi)^2 + \dots] dV, \quad (31)$$

where:

- $a(T) = a_0(T - T_c)$ changes sign at the critical point,

- $b > 0$ ensures thermodynamic stability,
- $\kappa > 0$ penalises density gradients (interface cost).

The equilibrium density profile minimises $F[\phi]$, yielding the Euler–Lagrange equation:

$$2a\phi + 4b\phi^3 - 2\kappa\nabla^2\phi = 0. \quad (32)$$

7 Monitoring Infrastructure

The gas2 module exposes a steady-state monitoring window displaying the 8-component potential decomposition and free-energy functional in real time.

7.1 Potential Decomposition Display

Eight channels of U , each updated per evaluation cycle:

Channel	Symbol	Description	Units
1	U_{bond}	Bond stretching	kJ/mol
2	U_{angle}	Angle bending	kJ/mol
3	U_{torsion}	Dihedral rotation	kJ/mol
4	U_{vdW}	Van der Waals	kJ/mol
5	U_{Coul}	Coulomb electrostatics	kJ/mol
6	U_{pol}	Polarization	kJ/mol
7	U_{many}	Many-body corrections	kJ/mol
8	U_{total}	Total potential	kJ/mol

7.2 Free-Energy Functional Display

The free-energy functional $F[\phi]$ is evaluated on a 1D density profile $\phi(x)$ across the simulation domain, with coefficients a , b , κ derived from the EOS:

$$a = \frac{1}{2} \left(\frac{\partial^2 A}{\partial V^2} \right)_{T,N}, \quad b \sim \frac{1}{4!} \left(\frac{\partial^4 A}{\partial V^4} \right)_{T,N}, \quad \kappa \propto \sigma^2 \varepsilon, \quad (33)$$

where σ and ε are the LJ parameters for the species.

8 Implementation Mapping

Theory	Code Location
$Z(T, V, N)$, Λ	include/gas2/gas2_thermo.hpp
$A(T, V, N)$, A^{dep}	include/gas2/gas2_thermo.hpp
$G(T, P, N)$	include/gas2/gas2_thermo.hpp
μ_i , ϕ_i , f_i	include/gas2/gas2_thermo.hpp
P_{sat} (Maxwell construction)	include/gas2/gas2_thermo.hpp
U decomposition	include/gas2/gas2_potential.hpp
$F[\phi]$ functional	include/gas2/gas2_potential.hpp
Monitoring window	tools/gas2_monitor.py
Tests	tests/test_gas2.cpp